

Molecular hydrogen in EXHALE

Abstract

EXHALE carries H_2 as a single species. The state vector holds one number density for it, the network reacts it as one reagent, and no rovibrational level is transported. Every quantity that depends on how the molecules are distributed over (v, J) therefore has that distribution supplied by assumption — and the assumption is not the same in every place. The infrared line cooling is explicitly LTE and says so in its own function name; the thermal dissociation and its three-body reverse rest on vibrational equilibrium and are now built from one another through the equilibrium constant, so that axis is consistent by construction; the Lyman–Werner self-shielding factor imports the non-LTE distribution of a 10^2 K interstellar photodissociation front into a collision-dominated base above 10^3 K; and the H_3^+ cooling, alone among the four, carries an explicit departure factor. This memo records each assumption, the published evidence behind it, the range over which it holds, and where the layer EXHALE actually computes leaves that range.

1. What is and is not carried

H_2 enters the state vector as species `isp_H2` of `src/modules/init/species_table.f90`, one of the four molecular columns (`H2`, `H2p`, `H3p`, `HeHp`) that the molecular option adds. It is written to `output/Ion_species.txt` and `output/Ion_species_adv.txt` as the number density of the molecule; the four molecular columns are the last four of the file, but their absolute position depends on which metal elements the run carries, so they must be located from the `# columns` schema header rather than counted.

What the state vector does *not* carry is any resolution of the rovibrational ladder. There is one H_2 continuity equation, not one per level, and no state-to-state collisional network. Nothing in the code transports a vibrational temperature, an ortho/para ratio, or a level population of any kind. The rovibrational ladder appears twice as *tabulated data* — the partition function of the equilibrium constant (`mol_rates.f90`) and the line list of the infrared cooling (`molecular_infrared_data.f90`) — but in both places it is evaluated at the local kinetic temperature and never solved for.

The consequence is the subject of this memo. A rate coefficient measured on one distribution of levels does not transfer to another, and EXHALE uses coefficients measured under at least three different distributions in the same cell. Each of them is defensible where it is used, and the point of collecting them is to say exactly where that is.

1.1 Code map

Table 1 lists every file that touches H_2 and what it does with it. Three groups are worth separating in advance, because the level-distribution assumption enters each of them differently: the files that supply a *rate*, those that supply an *emissivity*, and those that supply a *composition*.

1.2 The layer the assumptions are evaluated in, and what the numbers below are read from

Every claim about whether a fit is used inside or outside its range needs a statement of the conditions EXHALE actually reaches. The reference case here is `mol_lyman_werner` of the regression matrix — the hot-Uranus molecular base with

File	What it does with H ₂
init/	
species_table.f90	Declares isp_H2 and the nucleus weights (2 H nuclei per molecule) the element budget counts with
set_IC.f90	Seeds the H ₂ column from the chemical-equilibrium fit at the local pressure and T_0 , capped by the available neutral H
lower_atmosphere/	
mol_rates.f90	The Koskinen et al. (2022) Table-1 network, plus the H ₂ thermochemistry: partition function, equilibrium constant, three-body recombination
lower_column.f90	q_h2_equilibrium, the chemical-equilibrium mixing ratio, and the mean molecular weight of the analytic column built from it
lyman_werner.f90	Solomon-process photodissociation and the Draine & Bertoldi (1996) self-shielding function
molecular_infrared_cooling.f90, . . . _data.f90	The H ₂ quadrupole and magnetic-dipole line cooling, and the H ₂ O and CO bands beside it
h3p_cooling.f90	H ₃ ⁺ infrared cooling; H ₂ enters as the collider density of the non-LTE departure factor
nonlinear_system_solver/	
System_HeH_mol.f90,	The coupled ionization system: unknown $x(4) = 2n_{\text{H}_2}/n_{\text{H}}$, the H nuclei bound in H ₂
System_HeH_mol_metals.f90	
constrained_chemical_equilibrium.f90	The same network in logarithmic densities, with a continuation in the radiation field seeded from the H ₂ dissociation equilibrium
functions/	
composition.f90	h2_mixing_ratio_base, h2_bound_fraction, comp_ntot_bc: the H nuclei bound into H ₂ leave the base particle count
cross_sec.f90	sigma_H2, the photoionization cross section above 15.4 eV
binary_element_diffusion.f90	H ₂ as a diffusion carrier of the hydrogen component, with mass 2 and its own polarizability
radiation/ and post_process/	
util_ion_eq.f90	fuv_lw_photon_field: the shared 912–1110 Å beam, the H ₂ column, and the band fraction the pumping lines remove
Cool_coeff.f90	ioniz_HeI23S_H2, the He(2 ³ S) + H ₂ Penning and associative ionization rate
post_process_adv.f90	Does <i>not</i> carry H ₂ : the _adv reconstruction is molecule-free by construction, and says so in its header

Table 1: Where H₂ appears. Paths are relative to src/modules/.

Stellar LW flux [erg/cm²/s]: 343.0 — because it is the only default case that carries the Lyman–Werner channel. Measured on its output/Lyman_Werner.txt and output/Ion_species.txt:

Layer definition	cells	r/R_p	T [K]	n_H [cm ⁻³]
$x_{H_2} > 0.5$	227	0.9998–1.127	1149–1307	1.67×10^{12} – 9.27×10^{13}
$x_{H_2} > 0.01$	251	0.9998–1.172	945–1307	6.24×10^{11} – 9.27×10^{13}
$x_{H_2} > 10^{-6}$	—	0.9998–3.853	945–2430	1.98×10^7 – 9.27×10^{13}

with $x_{H_2} = 2n_{H_2}/n_H$; the H₂ density itself peaks at 4.63×10^{13} cm⁻³ in the base cell and falls to 4.42×10^9 cm⁻³ at the 1% edge. The three rows matter because they give three different answers. The often-quoted range 945–2430 K is the extent of gas that carries *any* H₂ at all, and its upper half is gas where H₂ is a 10⁻⁵ trace; the layer where H₂ is actually a constituent spans 945–1307 K. Statements below use the narrow definition unless they say otherwise.

These are relaxation snapshots, not converged solutions. Six of the eight default regression cases — every molecular one, plus lower_profile — are pinned at 12,000 steps by a maxsteps file and stop there rather than at a convergence criterion. mol_lyman_werner ends at $du = 2.24$, i.e. a 224% radial spread in the mass flux $\rho v r^2$, three orders of magnitude from the 10⁻³ threshold a converged run meets; only the two wasp cases converge. Every profile number in this memo therefore describes a relaxation snapshot. That is adequate for the purpose — the question is which decade of density and temperature a fit is being asked about, not the fourth digit — but it is not adequate for quoting a mass-loss rate, and none is quoted here.

And the snapshots do not currently match the goldens. At the time of writing, all eight default cases differ from backup/regression/golden/ in Hydro_ioniz.txt and Ion_species.txt. The working tree carries uncommitted source changes, including the one described in §2.2, and the goldens predate them. The numbers above are from the case output directories as they stand, not from the goldens.

2. The level-distribution assumptions

2.1 Infrared line cooling: LTE, stated explicitly

molecular_infrared_cooling.f90 evaluates the H₂ line emission as a Boltzmann sum,

$$E_{H_2}(T) = \sum_l f_u(T) A_l \Delta E_l, \quad f_u(T) = \frac{g_u e^{-T_u/T}}{Z(T)}, \quad Z(T) = \sum_{lev} g e^{-T_{lev}/T}, \quad (1)$$

and names the routine that returns it h2_line_emission_lte. There is no ambiguity about the assumption here; the question is only where it holds.

The line list is Roueff et al. (2019), A&A 630, A58, table 2 (electric quadrupole plus magnetic dipole), and the level ladder comes from the same table. As carried in molecular_infrared_data.f90 it is n_h2_lev = 302 levels and n_h2_line = 1833 lines.

Rotational levels are not the constraint: they thermalize at densities far below anything in this layer. The vibrational levels are. The code records the comparison at its own site: $n_{crit}(H_2 v = 1) \sim 10^9$ – 10^{11} cm⁻³ against 10^{13} – 10^{14} cm⁻³ at the base — and the measured base density of the molecular gate case, $n(H_2) = 4.63 \times 10^{13}$ cm⁻³, sits in that range.

Above the base the density falls, but so does the temperature, and the vibrational bands then carry a small share of the emission. That is the quantitative statement that makes the LTE overestimate harmless where LTE is least justified, and it can be evaluated directly from the line list: sum the emission of the lines whose upper level has $v \geq 1$ and divide by the total. Table 2 does so.

T [K]	$E_{\text{H}_2}(T)$ [erg s ⁻¹ molecule ⁻¹]	share from $v_u \geq 1$
400	3.034×10^{-23}	0.72%
500	7.717×10^{-23}	5.70%
600	1.789×10^{-22}	18.2%
1000	2.726×10^{-21}	65.7%

Table 2: Share of the LTE H₂ line emission carried by transitions out of a vibrationally excited upper level, evaluated on the implemented line list. Outside the tabulated range the share is 0.016% at 300 K and 81.5% at 2000 K. The level table carries no (v, J) index, so the vibrational assignment was recovered from the Roueff et al. (2019) source table and matched back onto the 1833 selected lines; classifying instead by $v_u \neq v_l$ (a band transition rather than a pure rotational one) changes the shares by less than 0.1 percentage point up to 1000 K.

The reading is that the LTE assumption is safe at the base, where the density supports it, and that where the density no longer supports it the vibrational bands — the levels the density fails to thermalize — have already stopped carrying the emission. Below about 500 K the error LTE can make is bounded by a few per cent of the total, because that is all the vibrational levels contribute.

2.2 Thermal dissociation and three-body recombination: vibrational equilibrium

The pair R12 ($\text{H}_2 + \text{M} \rightarrow \text{H} + \text{H} + \text{M}$) and R15 ($\text{H} + \text{H} + \text{M} \rightarrow \text{H}_2 + \text{M}$) is one channel run in two directions. Since the change recorded below, `mol_rates.f90` builds R12 from R15 through the equilibrium constant,

$$k_{\text{diss}}(T) = \frac{k_{\text{rec}}(T)}{K_{\text{eq}}(T)}, \quad K_{\text{eq}}(T) = \frac{n_{\text{H}_2}}{n_{\text{H}}^2} [\text{cm}^3], \quad (2)$$

rather than transcribing an independent dissociation fit.

That construction is what makes this axis consistent. K_{eq} is built in `h2_thermochemistry_init` from the translational partition functions and the *rovibrational* internal partition function `q_rovib_H2`, summed over the Morse ladder of H₂ $X^1\Sigma_g^+$ with the ortho/para nuclear-spin weights. It therefore contains vibrational excitation explicitly: both sides of the equilibrium refer to a vibrationally equilibrated gas, and the reverse rate built from it inherits that and nothing else.

The measured coefficient it is anchored on refers to the same state, and the source says so. Breshears & Bird (1973), 14th Symposium (International) on Combustion, 211, read their initial rate off the minimum of the postshock density-gradient dip, which they identify as “the rate corresponding to conditions of vibrational equilibrium in the thermal sense but with negligible extent of dissociation”. The same paper bounds where its rate law applies: the “observation of induction periods for diatomic dissociation clearly indicates that the assumptions implicit in Eqs. (2)–(4) are not valid during the very early course of the reaction; nevertheless, they may be applicable during the ‘steady state’ which follows the induction period.” A hydrodynamic model that resolves nothing faster than a vibrational relaxation time is

in that steady state throughout. Their Eq. (3), $k_d/k_r = K_{\text{eq}}$, states the detailed-balance relation outright, so deriving one direction from the other is the relation the measured coefficient already carries rather than an assumption added here. The critical density for vibrational thermalization, $10^4\text{--}10^8\text{ cm}^{-3}$ depending on the collider, sits many orders below the $10^{13}\text{--}10^{15}\text{ cm}^{-3}$ of the base, which is the independent check that the premise holds where the rate is evaluated.

Status of this change. R12 as detailed balance is present in the working tree and *not* in the last commit: `git show HEAD:src/modules/lower_atmosphere/mol_rates.f90` still carries the independent Baulch et al. (1992) fit $1.5 \times 10^{-9} e^{-48350/T}$, and `docs/Update_EXHALE.md` ends at section 114 with no entry for the change. The changelog is therefore behind the code on this point.

2.3 Lyman–Werner self-shielding: the weakest of the four

`lyman_werner.f90` takes the self-shielding factor from Draine & Bertoldi (1996), ApJ 468, 269 (hereafter DB96), their eq. (37),

$$f_{\text{shield}}(N_{\text{H}_2}) = \frac{0.965}{(1+x/b_5)^2} + \frac{0.035}{(1+x)^{1/2}} \exp\left[-8.5 \times 10^{-4}(1+x)^{1/2}\right], \quad x = \frac{N_{\text{H}_2}}{5 \times 10^{14}\text{ cm}^{-2}}, \quad (3)$$

with $b_5 = b/10^5\text{ cm s}^{-1}$ and b the thermal Doppler parameter $(2kT/m_{\text{H}_2})^{1/2}$.

This is where the imported level distribution is furthest from ours. DB96 built the fit for interstellar photodissociation regions: their stationary fronts run at $n_{\text{H}} = 10^2\text{ cm}^{-3}$ with the temperature parameter $T_0 = 200\text{ K}$ (their Figs. 3–6 and 8–11; Figs. 1 and 2 are 100 and 300 K LTE comparisons, and Fig. 7, the one this section leans on, is at $T = 10^2\text{ K}$ with no dust). In that gas the level distribution of the *shielded* H_2 is not thermal at all — it is set in non-LTE by ultraviolet pumping and by formation on grains. Neither T nor n_{H} is an argument of the fit (T enters only through b), so what those conditions really fix is the rovibrational population of the absorbing H_2 , which is the one ingredient of DB96’s calculation that cannot be rescaled from outside. Normalizing f_{shield} to the $N = 0$ rate carries with it the assumption that this distribution does not change with depth.

EXHALE evaluates it in a collision-dominated planetary base at 945–1307 K and n_{H} up to $9.3 \times 10^{13}\text{ cm}^{-3}$, eleven orders of magnitude denser than the fronts the fit was built on and five to six times hotter.

What partly resolves it, and what the paper actually says. DB96’s Fig. 7 compares the self-shielding produced by three different level distributions — a 100 K LTE population and two ultraviolet-pumped non-LTE ones at $\chi = 1$ and $\chi = 10$. Its *caption* states that “the three self-shielding factors are essentially identical for $N(\text{H}_2) \gtrsim 2 \times 10^{17}\text{ cm}^{-2}$ ”.

The *body* text of the same section is more careful, and it is the better ground to stand on:

“We see that for $10^{14} \lesssim N_2 \lesssim 10^{18}\text{ cm}^{-2}$, f_{shield} does show a dependence on the details of the H_2 rovibrational distribution. When there is higher excitation, the pumping occurs via a larger number of lines, so that the H_2 remains optically thin a bit longer, and self-shielding is less effective until one enters the heavily damped regime at $N_2 \gtrsim 10^{18}\text{ cm}^{-2}$. Overall, however, the self-shielding function $f_{\text{shield}}(N_2)$ is relatively insensitive to the effects of UV pumping with $\chi \lesssim 10^3$.”

So the authors do *not* claim distribution-independence at 10^{17} ; they claim a dependence up to 10^{18} and its disappearance in the heavily damped regime beyond. Our molecular layer sits at 10^{21} cm^{-2} and above, three decades inside that regime, which is a stronger position than the caption alone would give.

What is left. The insensitivity was demonstrated among *cold* distributions: a 100 K LTE population against UV-pumped ones, all at $T = 10^2 \text{ K}$ and $n_{\text{H}} = 10^2 \text{ cm}^{-3}$. None of the three is a 10^3 K collisionally equilibrated population, and no distribution at our temperature was tested. The physical argument for the damped regime — that once the pumping lines are on their damping wings the equivalent width stops caring which lines carry the absorption — is what makes the transfer plausible, not a demonstration. That is the honest statement of the position.

Two further axes. The *column* is where the layer leaves what was checked — though it is worth being precise about what that means, since DB96 put no upper bound on eq. (37) at all. Their §5.2 says only that it “does an excellent job in reproducing the initial rapid decline in f_{shield} for $10^{14} < N_2 < 10^{17} \text{ cm}^{-2}$, the ‘square root’ behavior for $10^{17} < N_2 < 10^{20} \text{ cm}^{-2}$, and the rapid falloff due to line overlap for $N_2 > 10^{20} \text{ cm}^{-2}$ ”. What *is* bounded is the comparison against exact multiline transfer, which in the same section reaches “the largest column densities considered, $N_2 = 3 \times 10^{21} \text{ cm}^{-2}$, where line overlap effects suppress the pumping rates by a factor ~ 10 ”.

The base cell of `mol_lyman_werner` settles at $N_{\text{H}_2} = 4.361 \times 10^{21} \text{ cm}^{-2}$ and passes through 5.31×10^{21} while relaxing, which the run reports once at the crossing. So the deepest cells use the fit 45% past the largest column at which anyone checked it — undemonstrated rather than outside a stated range, which is a weaker indictment but still a real one, since the regime beyond 10^{20} is exactly the line-overlap regime where the fit is doing its hardest work. There f_{shield} is 9.5×10^{-7} and falling as $N^{-3/4}$.

The *Doppler parameter* enters only through the saturated core, the $(1 + x/b_5)^{-2}$ term. Our thermal b is 2.79 km s^{-1} at 945 K and 4.48 km s^{-1} at 2430 K, inside the $1\text{--}10 \text{ km s}^{-1}$ span the form was built to cover. But at the columns of this layer the saturated term has collapsed — at $N_{\text{H}_2} = 4.4 \times 10^{21}$ it contributes $\sim 10^{-10}$ against the 9.5×10^{-7} of the damping-wing term — so f_{shield} is the same to seven digits at both b values. The b axis is neither tested nor, here, used: what the layer’s shielding actually rests on is the second term of eq. (37), the one with no b in it.

2.4 H_3^+ cooling: a departure factor, not LTE

The one place where EXHALE does not assume LTE is the H_3^+ infrared cooling, and it is worth stating because it runs the other way from §2.1. `h3p_cooling.f90` multiplies the Miller et al. (2013) LTE emission fit by the departure factor $s(T, n_{\text{H}_2})$ of their Table 6, bilinearly interpolated in $(T, \log_{10} n_{\text{H}_2})$, clamped at the table edges and set to 1 for $n_{\text{H}_2} \geq 10^{14} \text{ cm}^{-3}$. Here H_2 enters as the collider density that sets how far the H_3^+ levels depart from a Boltzmann distribution.

The net exchange is written in Kirchhoff form,

$$\Lambda_{\text{net}} = n_{\text{H}_3^+} 4\pi s(T, n_{\text{H}_2}) [E(T) - W E(T_{\text{rad}})], \quad (4)$$

with W the dilution of the field the unresolved lower atmosphere presents, switched on by the input key `Base IR field` (default `False`). The departure factor multiplies both terms, which it must: $s < 1$ applied to the emission alone would

heat a gas hotter than the field, and the form above is exactly zero at $T = T_{\text{rad}}$, $W = 1$.

The factor is doing real work rather than sitting at its LTE value: evaluated over the $x_{\text{H}_2} > 0.01$ layer of `mol_lyman_werner`, s runs from 0.996 in the base cell ($n_{\text{H}_2} = 4.63 \times 10^{13} \text{ cm}^{-3}$) down to 0.438 at the top of that layer ($4.42 \times 10^9 \text{ cm}^{-3}$, $r = 1.172 R_p$). Assuming LTE there would overstate the H_3^+ cooling by a factor 2.3.

The table is a vacuum departure factor — collisions against spontaneous decay, no incident field — so a field strong enough to matter pumps the levels back toward LTE and the true departure is smaller than s . The treatment therefore underestimates the magnitude of the exchange, not its sign or its fixed point.

For comparison, Frelikh & Murray-Clay (2026), ApJ 996, 96 treat the same coolant in pure LTE. On this one channel EXHALE carries the more complete physics.

2.5 The chemical-equilibrium mixing ratio

`q_h2_equilibrium` (`lower_column.f90`) is the Koskinen et al. (2022) Eq. (11) fit to the Visscher et al. (2006) chemical-equilibrium H_2/H partition,

$$q_{\text{H}_2} = \frac{1.9845 + 10^u - \sqrt{10^u(3.9690 + 10^u)}}{2.3670}, \quad u = -\frac{23672}{T} - \log_{10} p[\text{bar}] + 6.2645, \quad (5)$$

with limits $1.9845/2.3670 = 0.83840$ as $u \rightarrow -\infty$ (fully molecular, the fit's approximation of the solar-composition volume mixing ratio $0.5/(0.5 + \text{He}/\text{H}) = 0.863$) and 0 as $u \rightarrow +\infty$.

It is used in four places: the mean molecular weight of the analytic lower column, the base H_2 mixing ratio when `no_base.inp` supplies a photochemical one, the initial-condition seed of the H_2 column, and the $\lambda \rightarrow 0$ end of the continuation path in `constrained_chemical_equilibrium.f90`.

The assumption here is not about level populations but about the chemistry: this is *thermochemical* equilibrium, and it knows nothing about photodissociation. It is therefore not a statement about an irradiated wind, and the code says so at its own site — in chemical equilibrium $q_{\text{H}_2}(1 \mu\text{bar})$ stays large up to $T \sim 2000 \text{ K}$ and the gas is fully atomic only above $\sim 2400 \text{ K}$, while photochemistry plus the hotter actual base temperature dissociate H_2 well below that. Using it as a *seed* for a solve that then applies the full radiation field is exactly the use it supports; using it as a discriminant for whether a cell should be molecular is not. Section 114 of `docs/Update_EXHALE.md` records an attempt in that direction being withdrawn, together with the repair of an asymptotic guard in the same function that had been inverted since transcription and handed every cell below about 529 K a fully atomic state labelled “the dense molecular limit”.

2.6 Three sites that inherit a ground-state assumption silently

Three further quantities are measured on ground-state H_2 and used without a level-distribution correction. None is flagged in the code as such, and none is likely to matter at the level the three above do, but the list should be complete.

- **Photoionization.** `sigma_H2` in `functions/cross_sec.f90` is the Yan, Sadeghpour & Dalgarno (1998), ApJ 496, 1044 fit (their Eqs. 17–19), threshold 15.4 eV, with the near-threshold resonance structure smoothed as in the source fit. It is a cross section for H_2 in its ground vibrational state; a vibrationally hot molecule has a lower threshold and different near-edge structure.

- **Diffusion.** `binary_element_diffusion.f90` gives H₂ the static dipole polarizability $\alpha = 5.315 a_0^3$ (Schwerdtfeger & Nagle 2019) for the ion-neutral coefficient. This is the $v = 0$ value; polarizability rises with vibrational excitation.
- **Metastable-helium Penning ionization.** `ioniz_HeI23S_H2` in `radiation/Cool_coeff.f90` is the Maxwell–Boltzmann average of the Cohen & Lane (1977) cross sections, taken from the Garcia Munoz (2025) network file, valid over 500–10⁴ K following its tabulation. It is a thermal average over relative translational velocity with the internal state of the H₂ fixed.

3. The third body

R12, R13 and R15 are three-body reactions written as a two-body coefficient times a density. Koskinen et al. (2022) print that density as “*n*” and do not say which particles it counts; their models are H₂/H-dominated Jovian and Neptunian envelopes, where M is H₂ and H. EXHALE passes the total gas-particle density with electrons excluded (`ion_cell_state%ntot`), so every heavy particle is treated as an equally efficient third body.

The coefficient this assumes uniform is the M = H₂ one: R15 carries Cohen & Westberg’s (1983) recommended $k_1(\text{H}_2) = 2.8 \times 10^{-31} T^{-0.6} \text{ cm}^6 \text{ s}^{-1}$ over 50–5000 K, labelled M = H₂ on their data sheet and separate from their $k_1(\text{H})$.

Breshears & Bird measured all three of their colliders in one experiment with one analysis, so their dissociation coefficients are the clean statement of how much the third body matters:

$$\begin{aligned} k_d(\text{M} = \text{Ar}) &= 1.55 \times 10^{-10} e^{-44736/T}, \\ k_d(\text{M} = \text{H}_2) &= 5.48 \times 10^{-9} e^{-52989/T}, \quad [\text{cm}^3 \text{ s}^{-1}] \\ k_d(\text{M} = \text{H}) &= 3.52 \times 10^{-9} e^{-43881/T}. \end{aligned} \quad (6)$$

Evaluated here, $k_d(\text{H})/k_d(\text{H}_2)$ is 8.67 at 3500 K, 6.26 at 4000 K, 3.97 at 5000 K and 2.01 at 8000 K — a factor of a few across their measured 3500–8000 K range. Cohen & Westberg’s independent *recombination* pair, $k_1(\text{H}) = 8.8 \times 10^{-33}$ against $k_1(\text{H}_2) = 2.8 \times 10^{-31} T^{-0.6}$, gives 4.21 and 5.21 at 3500 and 5000 K, which is the agreement detailed balance requires of two independent sources inside their ranges.

Below the measured range the two Arrhenius forms diverge, because their activation temperatures differ by 9108 K: the ratio reaches 13.4 at 3000 K, 61.0 at 2000 K and 5.80×10^3 at 1000 K, against 1.98 from Cohen & Westberg at 1000 K. Take a factor of 2 to 5 as the collider spread where H₂ exists, and read the 5.8×10^3 as the cost of extrapolating an Arrhenius fit rather than as physics.

Helium is the collider nobody measured. The equal-efficiency assumption is weakest in two places, and only one of them is quantified above. Above the H₂ front atomic H dominates the collider population, and the factor of 2–5 is the size of the error there — carried, at least, by a vanishing H₂ density. In a helium-rich envelope M is helium, and there is no measurement to compare against at all. Breshears & Bird shocked H₂ in argon and in xenon (5.0, 10.0, 20.0% H₂ in Ar; 5.0, 10.0, 20.2% in Xe) and extracted M = Ar, Xe, H₂ and H; helium appears in their paper only as a shock-driver gas and in “helium–neon laser”. None of the sources behind R12, R13 or R15 supplies a helium efficiency, and none is applied. On general grounds a monatomic third body has no internal modes to take up the

released energy and should be *less* efficient than H_2 , so in a helium-dominated base these three rates are upper bounds; docs/molecular_chemistry_audit_he_rich.md §4.1 puts the size of a plausible factor at order 0.5 and records it as a guess rather than a measurement. Nothing in this repository supports a helium efficiency near unity, and no such number should be quoted from it.

4. Validity ranges

Table 3 collects every range in one place: what each treatment stands on, and what EXHALE asks of it. The evaluated column is the $x_{\text{H}_2} > 0.01$ layer of §1.2 unless stated, with the caveats recorded there about relaxation snapshots.

Table 3: What each treatment stands on and what it is asked to do.

Treatment	Range of the data	Range evaluated	Verdict
H_2 LTE line emission	Vibrational thermalization needs $n \gg n_{\text{crit}}(v=1) \sim 10^9 - 10^{11} \text{ cm}^{-3}$	$n(\text{H}_2) = 4.4 \times 10^9 - 4.6 \times 10^{13} \text{ cm}^{-3}$	Inside at the base; aloft the vibrational bands carry $< 6\%$ of the emission below 500 K, which bounds the error
H_2O and CO band cross sections	Tabulated 50–2000 K, clamped outside	945–1307 K in the molecular layer	Inside. Both molecules are largely dissociated above 2000 K in any case
R15 three-body recombination, $k_1(\text{H}_2) = 2.8 \times 10^{-31} T^{-0.6}$	Cohen & Westberg (1983), recommended 50–5000 K; ± 0.2 in $\log k$ at 300 K rising to ± 0.4 at 5000 K	945–1307 K	Inside
R12 thermal dissociation, $k_{\text{rec}}/K_{\text{eq}}$	Inherits R15's range; K_{eq} tabulated 100–20000 K and clamped	945–1307 K	Inside. Against the Baulch et al. (1992) fit the ratio is 0.97–1.18 over that fit's own 2500–8000 K and departs only below it
Third body M = everything	Measured for M = H_2 , H, Ar, Xe (Breshears & Bird 1973); no helium coefficient exists	Total heavy-particle density	H_2 -dominated base: inside. Above the front, M is atomic H, a factor 2–5. Helium: unquantified
Lyman–Werner f_{shield} , column axis	No upper bound stated. Checked against exact transfer to $3 \times 10^{21} \text{ cm}^{-2}$	up to 4.36×10^{21} , and 5.31×10^{21} in transient	Past what was checked , by 45% in column, in the line-overlap regime

Table 3 (continued)

Treatment	Range of the data	Range evaluated	Verdict
Lyman–Werner f_{shield} , Doppler axis	Every exact multiline check DB96 ran was at $b = 3 \text{ km s}^{-1}$; the 1–10 km s^{-1} axis is constructed	thermal $b = 2.79\text{--}4.48 \text{ km s}^{-1}$	Bracketing their one tested value, but the axis is untested. At our columns the b -dependent term is negligible anyway
Lyman–Werner f_{shield} , level distribution	$T = 10^2 \text{ K}$, $n_{\text{H}} = 10^2 \text{ cm}^{-3}$, non-LTE (UV pumping and grain formation); Fig. 7’s three distributions	945–1307 K, n_{H} up to $9.3 \times 10^{13} \text{ cm}^{-3}$, collisional	Outside. DB96 put the distribution dependence at $N_2 \lesssim 10^{18}$ and our columns are three decades above it, but no 10^3 K distribution was tested
Band fraction A from DB96 eq. (39)	Their own ceiling, §4.3 eq. (29), is $W_{\text{max}} \approx \ln(1110/912) \approx 0.2$, i.e. $A = 1$	asymptote 1.0843 at $b = 2.79, 1.1050$ at 4.48 km s^{-1}	Above the authors’ own cap by 8.4–10.5%; clamped at 1 in the code
H_3^+ emission fit $E(T)$	Miller et al. (2013) Table 5, piecewise 30–5000 K; frozen above	945–1307 K	Inside. The published segments do not join; each is used on its own range
H_3^+ departure factor $s(T, n_{\text{H}_2})$	Table 6 grid $T = 300\text{--}5000 \text{ K}$, $\log_{10} n_{\text{H}_2} = 6\text{--}14$; clamped, $s = 1$ above 10^{14}	$T = 945\text{--}1307 \text{ K}$, $n_{\text{H}_2} = 4.4 \times 10^9\text{--}$ 4.6×10^{13}	Inside; $s = 0.44\text{--}0.996$ across the layer
q_h2_equilibrium	Koskinen et al. (2022) Eq. (11), validated against NASA CEA at 1000–2500 K; thermochemical equilibrium only	Seed and base-composition use only	Inside as a seed. Not applicable to an irradiated wind: it has no photodissociation
sigma_H2 photoionization	Yan et al. (1998) Eqs. 17–19, $E \geq 15.4 \text{ eV}$, ground vibrational state, resonances smoothed	the full stellar spectrum above threshold	Inside in energy; the vibrational-state assumption is unquantified
He(2^3S) + H_2 ionization	Cohen & Lane (1977) via Garcia Munoz (2025), tabulated 500– 10^4 K	layer temperatures, evaluated unclamped outside	Inside over the molecular layer

5. Open items

- **No (ν, J) resolution, and why it has not been added.** Carrying the ladder as transported species would mean 302 further continuity equations, the 1833-line radiative network as a level-coupling matrix instead of a summed emissivity, and a state-to-state collisional set that the code does not have and that none of its current sources supply. Against that cost, the place it would most obviously buy something — the self-shielding — is the place DB96 say the level distribution stops mattering, at columns above $\sim 10^{18} \text{ cm}^{-2}$, three decades below where our layer sits. The trade is recorded here rather than settled.
- **Self-shielding at our temperature is untested.** The insensitivity of §2.3 was demonstrated among cold distributions. A calculation of f_{shield} on a 10^3 K collisionally equilibrated population, at the columns of Table 3, would settle whether the fit transfers; nothing in the literature consulted here does it.
- **An internal inconsistency in the published DB96 fit.** DB96 cap the overlap-corrected equivalent width of the pumping lines at $W_{\text{max}} \approx \ln(1110/912) \approx 0.2$ (their §4.3, eqs. 29–30), i.e. at a band fraction of exactly 1. Their closed-form integral, eq. (39), has the asymptote $1.05(1 + 0.0117b_5)$ as $N_{\text{H}_2} \rightarrow \infty$, which at our Doppler parameters is 1.0843 ($b = 2.79 \text{ km s}^{-1}$) to 1.1050 (4.48 km s^{-1}): 8.4–10.5% above their own ceiling. Their remark below eq. (39) is only that it “corresponds to an equivalent width $W \approx \Delta \ln \nu \approx 0.2$ in the limit $N_2 \rightarrow \infty$ ”. The overshoot is a property of the published fit, not of our use of it, and is the same whether the band fraction is taken from eq. (39) or from integrating f_{shield} on the grid. `fuv_lw_photon_field` clamps it at 1 so the transmission handed to the continuum absorbers of the same interval cannot go negative.
- **The `mol_ir_bands` regression case does not test what its name says.** It sets `Base IR field: True` and `Molecular IR bands: True` but leaves `Oxygen chemistry` off, as its `EXHALE_resolved.out` confirms, so there are no H_2O or CO carriers. Measured on `output/Cooling_breakdown.txt`, columns 13 and 14 are exactly zero in all 504 rows, while column 12, the H_2 line channel, is nonzero everywhere and peaks at $2.934 \times 10^{-7} \text{ erg cm}^{-3} \text{ s}^{-1}$ at $r = 1.0025 R_p$, $T = 1316 \text{ K}$. The zeros are structural rather than numerical: with $n(\text{H}_2\text{O}) = n(\text{CO}) = 0$ the two net-rate functions return 0 on their own guard. The case gates the H_2 line channel and nothing else, and a case that exercises the H_2O and CO paths would have to add `Oxygen chemistry: True`. *Note also* that the golden directory for this case carries only `Hydro_ioniz.txt`, `Ion_species.txt` and `run.log`; `Cooling_breakdown.txt` is not snapshotted, so the infrared columns are not under regression control at all.

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